

Predicting Marriage Compatibility Using TabTransformer

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Introduction

Marriage patterns reflect societal structures, and this project analyzes how personal attributes influence matchmaking using NYT wedding data. Inspired by Brown's sociological research and recent advances in tabular deep learning, we apply TabTransformer to predict partner profiles from individual traits—framed as a multi-output classification task (gender, age, education, etc.). This approach could reveal matching biases or power recommendation systems while advancing Transformer applications to social data.

Dataset

We use a dataset of 9,160 wedding announcements published in *The New York Times* between 2012 and 2023. The data was collected via the NYT Article Search API and processed using GPT-4 and rule-based methods to extract structured demographic features for each individual. Annotated and cleaned by Dr. Zhenchao Qian and research scientist Guixing Wei at Brown's Data Science Initiative, the dataset includes over 60 variables such as gender, age (in ranges like 25-30), occupation (standardized by level and field), and education level.

Model Architecture

The ImprovedTabTransformer enhances the standard architecture by

- LayerNorm-stabilized feature embeddings → better training stability
- CLS token for global context aggregation → richer feature mixing
- Deeper output heads with GELU activation → improved multi-task prediction

Accuracy Results

Attribute	Accuracy
Gender	83.6%
School Tier	53.4%
Education Level	43.8%
Occupation	36.6%
Age Group	21.2%

Attribute	Input	Predicted Partner
Gender	Female	Male
Age / Age Group	28	31
School Tier	Ivy League / Elite Private	Ivy League / Elite Private
Education Level	S2	S3
Occupation / Field of Study	Business and Financial Occupations	Business and Financial Occupations

Bias Discussion

Our project lies at the intersection of sociology, relationships, and data modeling. Matchmaking and partner preferences are deeply tied to societal structures like class, race, education, and geography. Using machine learning to model romantic compatibility risks reinforcing existing inequalities: if historical patterns reflect homogamy (marrying within one's social group), the model may reproduce rather than challenge these biases. Additionally, applying such models to real-world matchmaking tools could lead to algorithmic steering without critical reflection. Key concerns:

- Self-selection bias: Only couples who choose to submit announcements (and are aware of the opportunity) are included.
- Editorial bias: NYT editors curate published announcements, favoring those with elite education, prestigious professions, and East Coast ties.

